Structure-Aware Multiprototype Prompting Method for Remote Sensing Farmland Segmentation

Miaoxin Lai, Yapeng Gao, *Member, IEEE*, Hongxia Deng, Hong Gao, and Haifang Li

***Abstract*—The segment anything model (SAM) has demonstrated remarkable potential for general-purpose visual segmentation. However, adapting SAM for remote sensing farmland image segmentation typically requires extensive pixel-level annotations and carefully designed point- or bounding-box prompts. Moreover, SAM has a limited capability to capture high-frequency structural details, which makes it difficult to precisely localize the complex and irregular boundaries of farmland parcels. To address these issues, we propose a structure-aware multiprototype prompt learning (SMPPL) method for accurate farmland segmentation. First, a structure-aware encoder is designed to progressively enhance farmland structural features with the hierarchical semantic features extracted from the SAM model. Then, a semantic–structure coordination module (SSCM) is developed to align semantic and structural representations through a dual-branch state-space architecture, thereby enabling structure-guided semantic refinement. By integrating these disparate feature spaces, the SSCM is able to reinforce the spatial dependency modeling of farmland parcels. Finally, a multiprototype-guided prompt module (MPGPM) is designed to capture high intraclass variance and automate prompt generation. This mechanism can effectively eliminate the need for manual prompt engineering and enhance the model's perception of parcel boundaries and regional integrity. On the FarmSeg-VL dataset, SMPPL achieves an intersection over union (IoU) of 86.22% and a Dice score of 91.45%, outperforming current state-of-the-art (SOTA) methods. The method also maintains consistent performance on other public remote sensing datasets, demonstrating its superior segmentation accuracy and robust generalization ability.**

***Index Terms*—Prototype prompt, remote sensing farmland segmentation, segment anything model (SAM), structure-aware.**

I. Introduction

AGRICULTURE forms the foundation of human survival and socio-economic development, and its stable growth is essential for ensuring national food security [1]. As the primary spatial units for crop cultivation and harvesting, farmland parcels constitute the core elements of agricultural management and applications. With the rapid advancement of precision agriculture, efficient and accurate acquisition of farmland parcel information has become a critical prerequisite for crop monitoring, area estimation, and optimized resource management. However, traditional approaches for collecting farmland parcel information mainly rely on field surveys, which are time-consuming and labor-intensive. In recent years, remote sensing images have demonstrated great potential in farmland resource monitoring and management due to their wide coverage, short revisit cycles, and convenient data acquisition [2], [3]. Consequently, extracting farmland parcels from remote sensing images has become a key step toward the deployment of large-scale precision agriculture [4], [5].

In recent years, deep learning has been successfully applied to remote sensing image analysis, enabling more accurate extraction of farmland parcels [6]. Unlike traditional methods such as edge detection [7], [8] and region-based segmentation [9], [10], [11], deep learning models learn high-level features directly from raw data in an end-to-end manner, greatly improving segmentation accuracy. Early farmland segmentation studies mainly used convolutional neural networks (CNNs) [12], [13], [14]. CNNs efficiently capture local textures and contour features through convolution operations. However, their reliance on fixed convolution kernels limits generalization, especially in complex and diverse agricultural scenes. In contrast, Transformer-based models [15], [16], [17] use self-attention mechanisms to capture global context and long-range dependencies among large farmland parcels. Despite these advantages, such models tend to suffer from the loss of fine-grained details and incur high computational costs, which restrict their practical applicability. Recently, CNN–Transformer hybrid architectures [18], [19] have been widely applied to farmland semantic segmentation due to their ability to jointly model local details and global context. However, existing methods still face limitations in feature fusion between CNNs and Transformers, which often result in semantic inconsistency and inaccurate boundary extraction. Moreover, these methods are typically trained on domain-specific and relatively small-scale datasets, which further limits their generalization ability in complex farmland scenes.

To address this limitation, vision foundation models (VFMs), such as the segment anything model (SAM) [20], have recently attracted increasing attention in image segmentation. SAM can automatically identify objects within an image and generate segmentation masks by simply receiving user inputs such as points indicating the inclusion or exclusion of objects, or bounding-box annotations [21]. Benefiting from large-scale pretraining on the spatial attention (SA)-1B dataset, SAM exhibits strong feature generalization capability and

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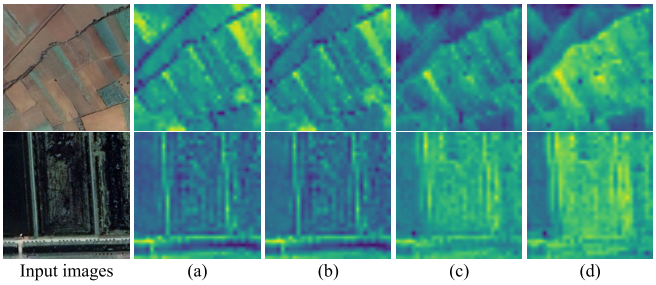


Fig. 1. Four-level features extracted by SAM. Low-level features (a) and (b) respond more strongly to local information, while high-level features (c) and (d) respond more strongly to global information.

robust semantic representation. Compared with CNN- and Transformer-based methods, SAM shows greater potential for handling complex farmland structures and varying spatial distributions. Motivated by SAM's outstanding performance on natural scene images, we focus on exploring its potential applications in remote sensing image segmentation.

However, due to the significant domain gap between natural images and remote sensing images, directly transferring SAM to remote sensing image segmentation often results in suboptimal performance [22], [23]. Several recent studies have attempted to adapt SAM from natural scene images to the remote sensing domain. Existing approaches primarily focus on parameter-efficient fine-tuning (PEFT) methods [24], [25]. These methods freeze the SAM encoder and insert learnable parameters to achieve adaptation in remote sensing segmentation tasks at a lower computational cost [26]. In addition, SAM heavily relies on manually annotated prompts such as bounding boxes and points to identify target objects. To address this limitation, some researchers have explored automatic prompt generation for SAM by designing more effective prompt strategies or introducing auxiliary prompt encoders [27], [28].

Although SAM has made preliminary progress in remote sensing segmentation, existing methods still face two major challenges. On the one hand, while the vision transformer (ViT) architecture effectively models semantic context, it lacks sufficient sensitivity to high-frequency structural features such as local edges and contours [29], as shown in Fig. 1. Current research on enhancing boundary segmentation within SAM remains limited, resulting in inadequate exploitation of the topological continuity and morphological integrity of farmland parcels [30]. On the other hand, farmland exhibits significant intraclass variance [31]. The morphological differences across various terrains are pronounced, and crops display distinct texture features at different growth stages. Lacking domain-specific prior knowledge about farmland, existing prompt mechanisms in SAM fail to fully capture the complex spectral-spatial feature patterns of diverse farmland objects, further constraining segmentation accuracy.

To address the challenges mentioned above, we propose a structure-aware multiprototype prompt learning (SMPPL) method for remote sensing farmland segmentation. Building upon the pretrained SAM, the method jointly models complex farmland structures and intraclass diversity through a dual-encoder feature modeling scheme and a prototype prompt mechanism. Specifically, we design a structure-aware encoder parallel to the SAM encoder, composed of four cascaded structural feature enhancement modules (SFEMs), which explicitly capture boundary and geometric topology information within farmland scenes. To fully leverage the complementarity between semantic and structural features, we further propose a semantic–structure coordination module (SSCM). Unlike traditional attention mechanisms that suffer from quadratic complexity when handling long sequences, state space models (SSMs) maintain strong sequence modeling capabilities with linear complexity. Thus, we incorporate state space modeling into the feature fusion task to obtain structure-aware joint feature representations.

Meanwhile, category-guided semantic segmentation methods [32] in remote sensing explicitly model class-level semantic priors and generate structured category-level prompts to constrain feature learning. Mechanistically, such category-guided segmentation aligns closely with the prompt-driven paradigm employed by SAM [33]. Motivated by this insight, we design a multiprototype-guided prompt module (MPGPM) for the SAM model. The module employs multiple learnable prototypes for prompt modeling, replacing explicit prompts based on points or bounding boxes. Through a mask-guided cross-attention mechanism, it dynamically integrates farmland prototypes with semantic features to generate latent prompt embeddings derived from category prototype learning. Finally, we validate the effectiveness of the proposed method on three publicly available farmland remote sensing datasets.

In summary, the contributions of this article are as follows.

1) We propose SMPPL for remote sensing farmland segmentation. By jointly modeling semantic features, structural information, and prototype prompts, our method achieves optimal performance in farmland parcel segmentation for high-resolution remote sensing images.
2) We design a semantic–structure coordination feature modeling mechanism. This architecture enhances geometric awareness of complex parcel boundaries through cascaded SFEMs and includes an SSCM module that can significantly improve the boundary and topological representation of farmland.
3) We introduce a prototype prompt mechanism to explicitly model the intraclass heterogeneity of farmland parcels. Through the cooperative optimization of a prototype adaptive optimizer and a dual-path prompt generator, the method provides dual-level guidance at both pixel and semantic levels to the SAM decoder.

The rest of this article is arranged as follows. Section II reviews the related work. Section III details the proposed method along with its key components. Section IV systematically evaluates the segmentation accuracy and generalization capability of the proposed method on farmland recognition tasks. Section V concludes this article and outlines future research directions.

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II. Related Work

A. Deep Learning-Based Farmland Segmentation Methods

In recent years, deep learning has been widely applied to semantic segmentation due to its powerful feature extraction capabilities, with its potential in farmland segmentation tasks continually expanding [34], [35]. Most existing deep learning-based farmland extraction methods follow traditional remote sensing segmentation strategies, primarily enhancing segmentation performance by capturing multiscale contextual information [36], [37], [38]. For example, Taravat et al. [39] proposed a ResU-Net architecture that integrates residual learning with the strengths of U-Net, specifically optimized for farmland boundary detection in Sentinel-2 imagery, providing a benchmark that balances accuracy and efficiency in this domain. To further address the extraction challenges posed by complex farmland scenarios such as saline-alkali lands in arid regions, Lu et al. [40] developed a fully convolutional network named DASFNet, which combines dual attention mechanisms with multiscale feature fusion, achieving high-precision automated farmland extraction in southern Xinjiang.

To balance global context and local details, multitask learning architectures have emerged as an effective approach to improve parcel-level extraction accuracy [41]. By jointly learning related tasks such as farmland mask extraction, boundary detection, and distance map estimation, these architectures enable feature sharing and mutual constraints between boundary information and semantic category features, effectively mitigating issues like blurred boundaries and omission of small fragmented parcels in complex scenarios [42]. For instance, Dong et al. [43] proposed DSFNet, which employs parallel branches for semantic segmentation and boundary detection, combined with morphological postprocessing to refine results. This approach achieves precise, large-scale farmland parcel extraction and exemplifies a semantic-boundary collaborative paradigm for parcel-level remote sensing segmentation. Beyond multitask learning, CNN–Transformer hybrid architectures have become one of the mainstream solutions for farmland semantic segmentation [44]. For instance, Liu et al. [18] introduced an edge-aware multitask network integrating CNN and Transformer architectures. Through a hard gating strategy that dynamically selects dual-channel features, their method enhances both holistic farmland recognition and boundary detail capture. Bu et al. [45] developed DCCLNet, which employs multiscale dual coordinate attention to specifically capture boundary-oriented features, effectively resolving the ambiguity in parcel boundaries.

To address the quadratic computational overhead of Transformers in high-resolution imagery, recent advancements have introduced SSMs such as Mamba. By achieving linear-complexity global context modeling, Mamba effectively balances computational efficiency and segmentation accuracy [46]. Wang et al. [47] proposed the PyramidMamba framework, which effectively resolves the issue of semantic redundancy in multiscale feature fusion by designing a dense spatial pyramid pooling (DSPP) and a pyramid fusion Mamba (PFM) module. However, such SAM–Mamba hybrids primarily utilize SSMs as efficient replacements for standard backbone components to fuse multiscale features. In contrast, the proposed SSCM tightly integrates high-frequency structural priors extracted from the encoder with the semantic representations of SAM, thereby establishing stronger spatial consistency.

B. Adapting SAM to Remote Sensing

Due to the significant domain gap between natural and remote sensing images, researchers often introduce lightweight modules or employ PEFT strategies to improve the adaptability of SAM in remote sensing tasks [48], [49], [50]. Among these, adapter-based methods insert lightweight adapters into the base model, updating only the adapter parameters while keeping the pretrained backbone frozen. For example, Chen et al. [24] proposed the SAM-Adapter framework, which injects task-specific knowledge by inserting lightweight adapters into Transformer layers, achieving promising results in camouflaged object detection and shadow detection tasks. Ye et al. [51] proposed a lightweight trident decoder network integrated with FastSAM, innovatively decoupling temporal-invariant semantic features from change features in bitemporal remote sensing images. Zhang et al. [52] proposed the RSAM-Seg framework, which adapts SAM for fully automatic remote sensing semantic segmentation by explicitly extracting multiscale boundary and texture priors and injecting them into the model to replace manual prompts. However, this adaptation scheme primarily relies on additive feature injection through adapters and entirely bypasses the native prompt mechanism.

Recent studies have focused on designing more effective prompt generation strategies or introducing additional prompt encoders to improve the adaptability of SAM. For example, Chen et al. [53] proposed RSPrompter, enabling class-specific instance segmentation for remote sensing images by using a prompt generator to produce continuous prompts instead of SAM's original discrete ones. Luo et al. [21] proposed SAM-RSIS, a two-stage progressive training framework that combines a ViT backbone with a box decoder to automatically generate object bounding-box prompts, eliminating the need for manual input. However, this model's performance heavily depends on the quality of the generated box prompts and involves a complex training process. Sultan et al. [28] proposed GeoSAM, which fine-tunes SAM for geographical segmentation by utilizing an external pretrained model to generate spatial point prompts and a large language model to provide semantic text prompts. While effective, such multimodal approaches heavily rely on the performance of external auxiliary models.

To alleviate the loss of high-frequency edge details in the ViT encoder of SAM, recent boundary-aware adaptations like boundary SAM [30] have been proposed. Boundary SAM enhances agricultural parcel delineation by exploiting SAM's image embeddings for PCA-based dimensionality reduction and detail-enhancement filtering. However, it mainly serves as an external enhancement pipeline and lacks semantic–structural interaction modeling and adaptive prompt learning, which limits its ability to fully exploit the representation and prompt-driven capabilities of SAM. Although these methods have improved the applicability of SAM in remote

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TABLE I
COMPARISON OF DIFFERENT METHODS IN SEGMENTATION TASKS

Method	Structure Modeling	Automatic Prompting	Feature interaction scheme
SAM-Adapter [24]	×	×	Adapter Injection
Pyramid-Mamba [47]	×	×	Mamba multi-scale fusion
RSAM-Seg [52]	×	✓	Multi-scale feature prior
GeoSAM [28]	×	✓	Multi-modal prompting
PGP-SAM [57]	×	✓	Prototype Driven
Boundary SAM [30]	✓	×	Shallow edge filtering
PDSSNet [31]	✓	×	Prototype Driven
SMPPL (Ours)	✓	✓	Semantic-structure coordination

sensing tasks to some extent, most existing approaches remain limited to fine-tuning local model modules. They struggle to fully capture the complex multiscale structures and significant intraclass variations in remote sensing scenes.

C. Class-Guided Semantic Segmentation of Remote Sensing Images

Unlike traditional context modeling methods, category-guided semantic segmentation explicitly incorporates class semantic information into the model to provide clearer guidance for feature extraction and optimization, thereby enhancing feature discrimination [54]. The current mainstream category-guided approaches utilize coarse segmentation maps generated from intermediate network layers as class priors. For example, Ni et al. [32] proposed CGGLNet to adaptively capture multiscale local details. Specifically, it produces coarse segmentation results at different network stages to explicitly guide global feature interactions with class information. CAGNet [55] models explicit pixelwise class differences through category attention, combining local multiscale feature extraction with global–local class guidance. Another category of methods improves segmentation performance by modeling object-level class context. For example, LoG-CAN++ [56] employs a collaborative design of global class-aware modeling and local class-aware connections, where local class centers act as intermediate perception elements linking pixels to global class centers. It innovatively introduces an affine transformation block to adapt to scale and orientation variations, suppress background noise, and reduce intraclass variance. Essentially, category prototypes and coarse segmentation priors can be reformulated as a higher order structured prompt signal. Integrating such priors seamlessly into the prompt framework of SAM enables class-aware mask prediction. Regarding prototype-driven strategies [27], [57], recent works like PDSSNet [31] have demonstrated the effectiveness of utilizing class prototypes for structural guidance in remote sensing segmentation. PDSSNet synergizes semantic prototypes with structural features to refine object boundaries within a standard encoder–decoder framework. However, such methods are inherently tailored for traditional supervised learning and lack the flexibility to adapt to foundation models.

As summarized in Table I, existing methods mainly focus on a single aspect, such as efficient feature interaction, prompt generation, or structure modeling. In contrast, the proposed method establishes a synergistic semantic–structure–prompt coordination framework. Specifically, the structure-aware encoder progressively extracts and enhances high-frequency geometric features from SAM's hierarchical semantic representations; the SSCM performs structure-guided semantic refinement through dual-branch state-space interaction with linear computational complexity, and the MPGPM further reformulates category prototypes as dynamic structured prompt embeddings for automatic mask generation. Through this unified collaborative design, the proposed method simultaneously preserves semantic consistency, structural continuity, and prompt adaptability for complex farmland segmentation scenarios.

III. METHODOLOGY

A. Overall Network Architecture

This article proposes SMPPL for remote sensing farmland segmentation, as shown in Fig. 2. The method is based on the pretrained SAM and uses a structure-aware encoder and a prototype prompt mechanism to jointly model complex farmland structures and intraclass diversity. Specifically, the input remote sensing image is fed into two parallel encoders. One is the SAM encoder pretrained on large-scale data. During training, the backbone weights are kept frozen, and only lightweight adapter modules are fine-tuned. The other is a structure-aware encoder designed in this study. It consists of four cascaded SFEMs that explicitly model boundaries and geometric topology in farmland scenes. Then, the SSCM is used to align and fuse features from both encoders, thereby obtaining structure-aware joint representations. To capture significant intraclass variability in farmland categories, the MPGPM is introduced to utilize multiple learnable prototypes for prompt modeling. Specifically, a prototype adaptive optimizer collaborates with a generation module to produce both sparse and dense prompt embeddings. These generated prompt embeddings replace the manually specified point, box, or mask prompts required in the original SAM pipeline, enabling automatic prompt generation for farmland segmentation.

In terms of the decoder design, we retain the lightweight two-way Transformer architecture from the original SAM mask decoder while enabling all of its parameters to be trainable. To fully exploit SAM's native decoding mechanism, we further propose an iterative self-prompting segmentation strategy. First, the fused image features are fed into the SAM decoder without any external prompts to generate an initial coarse mask. Subsequently, this coarse mask is utilized as a spatial prior by the MPGPM to produce both sparse and dense prompt embeddings. Finally, the fused image features and prompt embeddings are refed into the parameter-shared SAM decoder for refined mask prediction. During refinement, the sparse prompt embeddings are concatenated with SAM's default output tokens and interact with image features through the two-way Transformer, while the dense prompt embeddings are added to the fused image features to introduce fine-grained structural guidance. Through this coarse-to-fine self-iteration mechanism, the proposed framework achieves high-precision segmentation without manual prompts.

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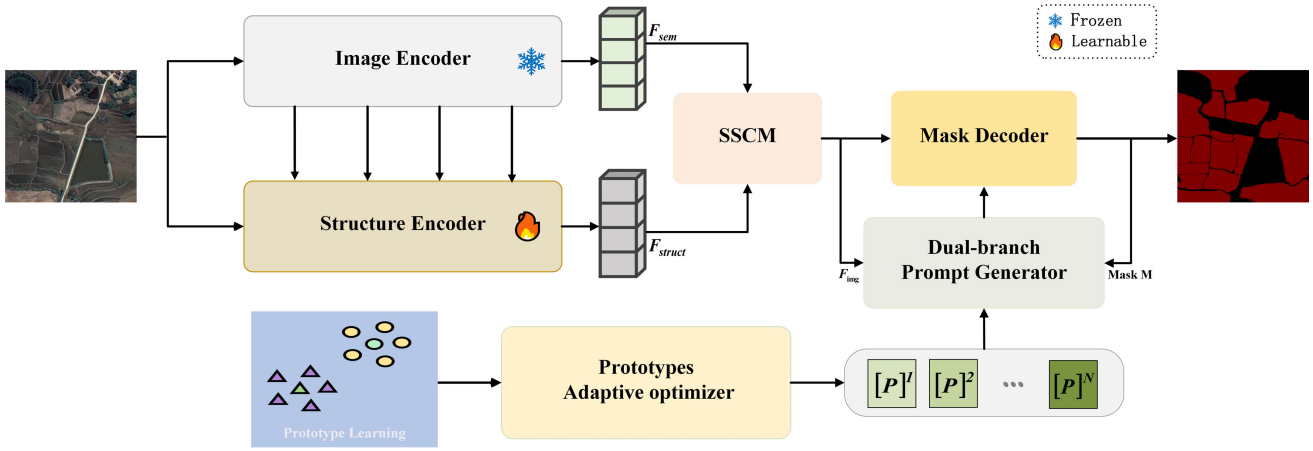


Fig. 2. Overview of the proposed SMPPL.

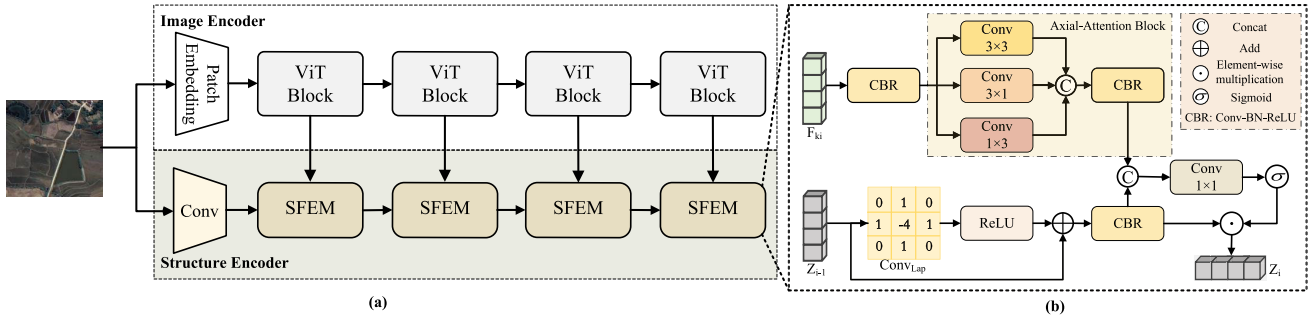


Fig. 3. Architecture of the proposed dual-stream encoder and the SFEM. (a) Architecture consists of an image encoder and a parallel structure-aware encoder. Hierarchical semantic features from the ViT blocks are progressively integrated into the structure-aware encoder through a sequence of SFEMs. (b) Detailed structure of the SFEM.

B. Structure-Aware Encoder

We propose a parallel structure-aware encoder branch composed of four cascaded SFEMs, as shown in Fig. 3. The input feature X first passes through a lightweight multibranch convolution block. This block uses parallel convolutions with different dilation rates to capture multiscale local structures. After the block, the structure branch enhances features layer by layer through the four SFEM modules. This forms a cascaded structural modeling process aligned layerwise with the semantic branch of the SAM encoder. For the ith SFEM module, its input consists of two parts: the output Z_{i-1} from the previous stage of the structure branch and the semantic feature F_{k_i} from the k_ith ViT block of the SAM image encoder. The computation of the ith SFEM module is thus given by the following equation:

$$Z_i = \mathrm{SFEM}_i\left(Z_{i-1}, F_{k_i}\right), \quad k_i \in \{3, 6, 9, 12\}. \tag{1}$$

The structure-aware encoder branch progressively incorporates high-level semantic guidance at different layers, producing the output F_{struct}. This enables a gradual fusion process from local boundary awareness to global structural modeling.

1) Structural Feature Enhancement Module: The proposed the SFEM progressively extracts and enhances high-frequency structural features from SAM's hierarchical semantic features. Rather than relying on explicit edge constraints or shallow filtering operations, SFEM captures intrinsic geometric structures across feature hierarchies, thereby enabling structure-aware feature refinement and improving spatial continuity in complex farmland parcels.

Let the output of the $(i-1)$th layer in the structure branch be Z_{i-1}. First, edge responses are extracted using a Laplacian convolution, followed by a ReLU activation to retain positive edge information. Then, a residual connection is introduced to obtain the enhanced edge feature S_i

$$S_i = Z_{i-1} + \mathrm{ReLU}\left(\mathrm{Conv}_{\mathrm{Lap}}\left(Z_{i-1}\right)\right) \tag{2}$$

where $\mathrm{Conv}_{\mathrm{Lap}}$ denotes a convolution operation based on a fixed Laplacian kernel. This process explicitly enhances the feature response of parcel boundary regions while preserving the original structural information.

To achieve effective spatial alignment between the semantic feature F_{k_i} from the semantic branch and the structural feature S_i from the structure branch, this article designs a lightweight axial attention block. The block models contextual dependencies along the horizontal, vertical, and local spatial directions, effectively capturing common strip-like and block-like structural patterns in farmland.

First, the input semantic feature F_{k_i} undergoes channel mapping and axial modeling. Specifically, the axial attention block employs three groups of parallel depthwise separable convolutions to extract multidirectional features, which capture long-range dependencies along the horizontal direction, structural variations along the vertical direction, and local spatial

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textures, respectively. Subsequently, the three axial features are concatenated with the original feature and fused to obtain a structure-aware feature representation V_i

$$V_i = \text{AxialAttention}\left(\text{CBR}\left(F_{k_i}\right)\right) \tag{3}$$

where CBR denotes a Conv-BN-ReLU convolutional block. After obtaining the edge-enhanced feature S_i and the structurally aligned semantic feature V_i, a SA map is further constructed to adaptively emphasize informative structural regions

$$Z_i = \sigma\left(\text{Conv}_{1\times1}\left(\text{Concat}\left(S_i, V_i\right)\right)\right) \odot S_i \tag{4}$$

where $\sigma(\cdot)$ denotes the Sigmoid activation function, $\odot$ denotes elementwise multiplication.

2) Semantic–Structure Coordination Module: Remote sensing farmland segmentation relies on both global semantic consistency and local structural boundary awareness. Existing studies mainly enhance global feature interaction by generating class-aware weights or employing self-attention mechanisms. However, the former often relies on global statistics and thus loses spatial details, while the latter suffers from high computational complexity and tends to cause oversmoothing in visually similar regions. As a result, it is difficult to precisely capture local structures, such as farmland boundaries, while maintaining semantic consistency. To address these limitations, SSCM is proposed to perform structure-guided semantic refinement through dual-branch state modeling and cross-branch state propagation.

We employ spatial-Mamba [58] to model spatial dependencies in feature representations. The model introduces a structure-aware state fusion (SASF) mechanism that establishes spatial interactions within the state space, enabling joint modeling of sequential dependencies and spatial structures. Given an input feature $U \in \mathbb{R}^{B\times H\times W\times C}$, the spatial-SSM module processes the feature through the following three equations:

$$x_t = Ax_{t-1} + BU_t \tag{5}$$

$$h_t = \sum_{k\in\Omega} \alpha_k x_{\varphi_k(t)} \tag{6}$$

$$y_t = Ch_t + DU_t. \tag{7}$$

Equation (5) represents the state transition equation, (6) denotes the SASF equation, and (7) corresponds to the output equation. Specifically, $A \in \mathbb{C}^{N\times N}$ is the state transition matrix, $B \in \mathbb{C}^N$ denotes the input control matrix, $C \in \mathbb{C}^N$ is the output projection matrix, and $D \in \mathbb{C}^1$ represents the skip-connection matrix. Ω denotes the spatial neighborhood set, α_k is a learnable fusion weight, and $\varphi_k(t)$ indicates the index of the kth neighbor at spatial position t.

In practice, multiscale dilated convolutions are employed to efficiently construct neighborhood connections and realize the SASF mechanism

$$h_t = \sum_{d\in\{1,3,5\}} \sum_{(i,j)\in\Omega_d} \text{K}_{ij}^d \cdot X_{t+i\cdot w+j} \tag{8}$$

where $\Omega_d = \{(i,j) \mid i,j \in \{-d, 0, d\}\}$ denotes the neighborhood set with dilation rate d, and K_{ij}^d represents the convolution kernel weight at the corresponding position. W denotes the feature map width, and the defined neighborhood states are used for state fusion.

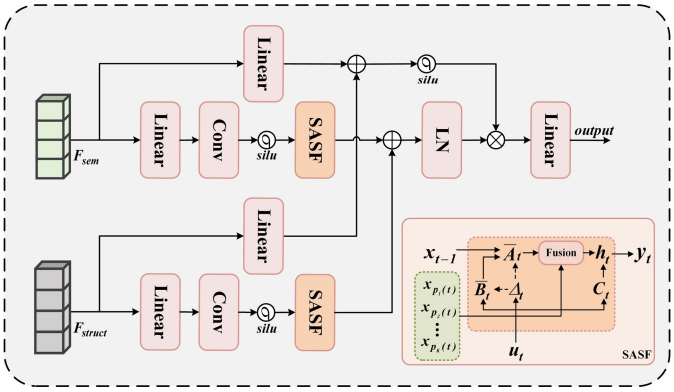


Fig. 4. SSCM.

The spatial-SSM module in spatial-Mamba is capable of explicitly modeling spatial dependencies within the feature domain. Building upon this state-space formulation, we innovatively integrate it into a dual-branch collaborative framework for joint semantic and structural modeling, as shown in Fig. 4. Given the semantic and structural features extracted from the encoder, $F_{\text{sem}}, F_{\text{struct}} \in \mathbb{R}^{B\times H\times W\times C}$, the 2-D feature maps are first flattened into sequence representations. The semantic sequence F_{sem} and the structural sequence F_{struct} are then fed into two independent Spatial-SSM branches to capture global contextual dependencies.

In our design, the semantic and structural branches share the output projection matrix C. Unlike independent output transformations, the shared projection introduces an implicit feature-space alignment constraint that improves cross-branch representation consistency in the final embedding space. It is worth noting that this shared projection mechanism is applied only at the output stage, while both branches still preserve independent state-space propagation processes and specific feature representations. The overall process can be formulated as follows:

$$F'_{\text{sem}} = \text{Spatial-SSM}_{\text{sharedC}}\left(F_{\text{sem}}\right) \tag{9}$$

$$F'_{\text{struct}} = \text{Spatial-SSM}_{\text{sharedC}}\left(F_{\text{struct}}\right). \tag{10}$$

The subscript sharedC emphasizes that the two branches share the projection matrix C. After spatial-SSM modeling, the two outputs are denoted as F'_{sem} and F'_{struct}. A concatenation-based fusion strategy is applied to build the coordinated representation

$$F_{\text{fused}} = \text{LN}\left(\text{Concat}\left(F'_{\text{sem}}, F'_{\text{struct}}\right)\right) \tag{11}$$

$$F_{\text{img}} = \text{Linear}\left(F_{\text{fused}} \odot \xi\left(\text{Concat}\left(F_{\text{sem}}, F_{\text{struct}}\right)\right)\right) \tag{12}$$

where Concat denotes concatenation, $\odot$ denotes elementwise multiplication, and $\xi(\cdot)$ is the SiLU activation function.

C. Multiprototype-Guided Prompt Module

Farmland categories in remote sensing images exhibit significant internal diversity, characterized by variations in parcel

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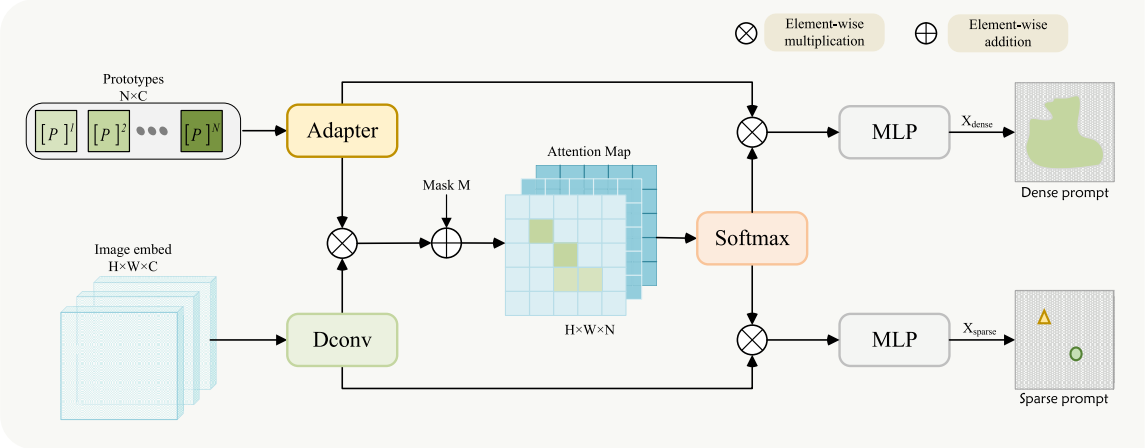


Fig. 5. Proposed dual-path prompt generator.

scale, complex texture patterns, and diverse spectral distributions. While recent SAM adaptation studies have explored automated prompt generation to replace manual annotations, these existing methods are suboptimal for complex agricultural scenes. Different from conventional prompt-generation methods, MPGPM formulates prompt generation as a prototype-guided semantic–structure representation process. Instead of generating a single prompt, MPGPM maintains a set of learnable global subprototypes to explicitly model the structural distribution of diverse farmland patterns. Furthermore, through the prototype adaptive optimizer and the dual-path prompt generator, these subprototypes are dynamically transformed into both dense spatial masks and sparse semantic embeddings. The prompt embeddings generated by this module directly fulfill the input requirements of SAM's decoder, significantly enhancing the model's discriminative capability in farmland scenes.

1) Prototype Adaptive Optimizer: To characterize the complex and diverse structural distributions within a single farmland category, we introduce multiple learnable global prototype vectors. Specifically, N farmland subprototypes are defined as follows:

$$P = \{p_1, p_2, \ldots, p_n\}, \quad p_i \in \mathbb{R}^c \tag{13}$$

where C denotes the feature dimension. In our experiments, the number of prototypes is set to $N = 5$. This optimal value is determined through parameter ablation experiments. All prototypes are initialized as learnable parameters and jointly optimized through gradient backpropagation during training. To enhance the adaptability of prototypes to the input image, a classwise attention-based dynamic prototype update mechanism is designed. Given the image feature X_{img} and the initial prototype set P, the prototypes are contextually updated via a multihead attention mechanism

$$P' = \text{Attn}\left(P, X_{\text{img}}, X_{\text{img}}\right) \tag{14}$$

where prototypes serve as queries, and image features act as key-value pairs. This process enables each subprototype to focus on spatial regions most relevant to its semantics, effectively transforming the prototypes from global static representations into image-adaptive semantic prompts, thereby improving their discriminative power and stability.

After obtaining the image-adaptive prototype set P', we further design a prompt generation module to convert the prototype information into prompt embeddings directly usable by the SAM mask decoder. Considering that the original SAM design supports both dense prompts and sparse prompts, this work implements a dual-path prompt generator to produce both dense and sparse prompts, as shown in Fig. 5. Specifically, the dense prompt branch focuses on preserving the spatial continuity of farmland parcels, while the sparse prompt branch emphasizes globally discriminative semantic information and region localization for reducing intraclass variability in complex agricultural scenes. Therefore, the two prompt generators follow different generation pipelines according to their distinct representation objectives, jointly enabling geometry-aware and semantically discriminative prompt adaptation.

2) Dense Prompt Generator: The dense prompt generator (DPG) aims to produce spatially aligned dense prompt feature maps while preserving the full spatial resolution of the input features. Given the image embedding $X_{\text{img}} \in \mathbb{R}^{B \times H \times W \times C}$ and the prototype set $P' \in \mathbb{R}^{N \times C}$, the DPG employs a prototype-guided attention mechanism combined with deformable convolution to effectively fuse semantic and spatial features.

Specifically, the input feature X_{img} is first processed by a deformable attention module for adaptive spatial feature extraction. Given the prototype set P', a learnable gating mechanism is initially applied to filter the prototypes

$$X_d = \text{DeformAttn}\left(X_{\text{img}}\right) \tag{15}$$

$$P_{\text{t}} = P' \odot \sigma(g), \quad g \in \mathbb{R}^c \tag{16}$$

where $\sigma(\cdot)$ denotes the Sigmoid activation function, and g represents the learnable gating parameter. In practical scenarios, a single remote sensing image rarely contains all structural variations of farmland. The gating mechanism $\sigma(g)$ dynamically evaluates the relevance of each prototype to the current image, suppressing irrelevant semantic prototypes and

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amplifying the active ones. This effectively prevents noisy or redundant information from interfering with the generation of dense prompts.

After obtaining the deformable attention features X_d and the enhanced prototypes P_t, we employ a cross-attention mechanism to enable deep interaction between semantic prototypes and spatial features. Specifically, the similarity matrix between the deformable features and prototypes is first computed and passed through a Softmax function to obtain the attention distribution of each spatial position over the prototypes. Finally, the prototype information is aggregated using the attention weights to generate the prototype-guided dense embedding X_{Dense}

$$X_{\text{Dense}} = \text{MLP}\left(\text{Softmax}\left(\frac{X_d P_t^T}{\sqrt{C}}\right) P_t\right) \tag{17}$$

where $\sqrt{C}$ serves as a scaling factor; the MLP consists of two convolutional layers with GELU activation functions. The generated dense prompt X_{Dense} is further integrated into the fused image feature to enhance spatial guidance for mask decoding. The integration process is formulated as follows:

$$\tilde{F}_{\text{img}} = F_{\text{img}} + X_{\text{Dense}}. \tag{18}$$

Through this operation, prototype-guided structural information is introduced into the spatial feature representation, thereby strengthening the decoder's ability to delineate farmland boundaries and maintain regional integrity.

3) Sparse Prompt Generator: In addition to dense prompts, to further construct discriminative global prompt representations, we propose a sparse prompt generator (SPG). Similar to the DPG, given the image embedding $X_{\text{img}} \in \mathbb{R}^{B \times H \times W \times C}$, the prototype set $P' \in \mathbb{R}^{N \times C}$, and the coarse mask features M, the SPG aggregates the image features into the prototype space through a masked attention mechanism, ultimately generating the sparse prompts X_{sparse}.

Specifically, dilated convolutions are first applied to the input feature X_{img} to enhance local context. Subsequently, the similarity matrix between the feature $\tilde{X}$ and the prototypes P' is computed and coupled with learnable gating weights $\sigma(g)$ to produce an initial attention distribution. Finally, a two-layer nonlinear projection is utilized to transform the weighted prototypes, enabling the model to dynamically activate more discriminative prototype representations P_a based on the semantic features

$$\tilde{X} = \text{DConv}\left(X_{\text{img}}\right) \tag{19}$$

$$w = \frac{\text{Softmax}\left(\text{Sim}\left(\tilde{X}, P'\right) \odot \sigma(g)\right)}{\sum_{j=1}^{N}\left[\text{Softmax}\left(\text{Sim}\left(\tilde{X}, P'\right) \odot \sigma(g)\right)\right]} \tag{20}$$

$$P_a = \text{MLP}\left(w \cdot P'\right) \tag{21}$$

where $\text{Sim}(\cdot)$ denotes the cosine similarity function, and $\sigma(g)$ is the learnable gating weight. After obtaining the updated sparse prototype set P_a, the sparse prototype-guided embedding X_{sparse} is generated by interacting with the image features through a mask-attention mechanism

$$X_{\text{sparse}} = \text{MLP}\left(\left(\text{Softmax}\left(\frac{P_a \tilde{X}^T}{\sqrt{C}}\right) + M\right)\tilde{X}\right). \tag{22}$$

Consistent with the prompt decoding process of SAM, the generated sparse prompt X_{sparse} is incorporated as additional latent query tokens. Specifically, X_{sparse} is concatenated with the decoder's default output tokens to form the token sequence for mask decoding. This token sequence is then fed into the two-way Transformer together with the prompt-enhanced image feature $\tilde{F}_{\text{img}}$, thereby providing semantic and localization information for farmland mask prediction.

D. Loss Function

To effectively enhance regional consistency and boundary structure preservation in farmland segmentation, this work adopts a joint loss function composed of binary cross-entropy (BCE) loss, intersection over union (IoU) loss, and boundary difference-overunion (BDoU) [59] loss. The BCE loss provides pixel-level supervision, constraining the consistency between predicted probabilities and ground truth labels. It is defined as follows:

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[g_i \log\left(p_i\right) + \left(1 - g_i\right)\log\left(1 - p_i\right)\right] \tag{23}$$

where p_i and g_i represent the predicted value and ground truth label of the ith pixel, respectively, and N denotes the total number of pixels. To enhance the model's ability to capture overall region overlap, the IoU loss is introduced, defined as follows:

$$\mathcal{L}_{IoU} = 1 - \frac{\sum_{i=1}^{N} p_i g_i}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} g_i - \sum_{i=1}^{N} p_i g_i}. \tag{24}$$

To enhance the model's ability to capture farmland boundary structures, this article introduces BDoU loss for optimization. Unlike the traditional Dice loss, which imposes a uniform constraint on overall region consistency, this loss dynamically adjusts the weighting by explicitly incorporating boundary complexity information, enabling adaptive modeling of structural regions. Furthermore, a squared term is integrated to intensify constraints on high-confidence predictions, further improving structural consistency. It is defined as follows:

$$\mathcal{L}_{BDoU} = \frac{\sum_{i=1}^{N} p_i^2 + \sum_{i=1}^{N} g_i^2 - 2\sum_{i=1}^{N} p_i g_i}{\sum_{i=1}^{N} p_i^2 + \sum_{i=1}^{N} g_i^2 - (1+\alpha)\sum_{i=1}^{N} p_i g_i} \tag{25}$$

where α is an adaptive weighting coefficient that dynamically adjusts according to the proportion of the target boundary

$$\alpha = 1 - 2\frac{L}{S} \tag{26}$$

where L denotes the boundary length of the ground truth parcel and S denotes its total area. Larger parcels yield a larger α to emphasize boundary optimization, while smaller or fragmented parcels produce a smaller α to balance region consistency and boundary preservation.

Combining the three losses described above, a weighted sum is adopted as the final optimization objective

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{BCE} + \lambda_2 \mathcal{L}_{IoU} + \lambda_3 \mathcal{L}_{BDoU} \tag{27}$$

where λ_1, λ_2, and λ_3 are the loss weighting coefficients. $\mathcal{L}_{BCE}$ and $\mathcal{L}_{IoU}$ are used as the main region-level supervision terms,

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TABLE II
DATASET DETAILS

Dataset Properties	Dataset Name		
	FarmSeg-VL	FGFD	LoveDA
Categories	2	2	7
Number	22605	2606	2358/5987
Spatial Resolution	0.5 m	0.3 m	0.3 m
Image size	512×512	512×512	1024×1024
Farmland Proportion	100%	100%	26.79%
Region	Eight Typical Agricultural Regions of China	East, South, West, North, Central (China)	Nanjing, Changzhou, Wuhan (China)

with their coefficients fixed at 1.0. $\mathcal{L}_{BDoU}$ is introduced as a boundary-aware constraint for contour refinement, and its coefficient is set to 0.5 based on sensitivity analysis.

IV. EXPERIMENTS

A. Datasets

For model evaluation, the FarmSeg-VL [60], FGFD [61], and LoveDA [62] datasets are adopted as benchmark datasets for remote sensing farmland semantic segmentation, and detailed statistics of these datasets are summarized in Table II.

1) FarmSeg-VL Dataset: FarmSeg-VL is the first fine-grained image–text dataset specifically designed for farmland segmentation. It covers eight representative agricultural regions in China, including the Huang–Huai–Hai Plain, Loess Plateau, Northeast Plain, northern arid and semiarid regions, Sichuan Basin, South China, the middle and lower reaches of the Yangtze River Plain, and the Yunnan–Guizhou Plateau, spanning all four seasons to ensure spatiotemporal balance. Each image has dimensions of 512×512 pixels and is annotated at the pixel level to distinguish farmland from nonfarmland regions. The dataset contains a total of 22 605 image–text–mask samples, which are split into training, validation, and test sets with a ratio of 7:2:1, corresponding to 15 821, 4512, and 2272 samples, respectively. In this study, only the image and mask data from FarmSeg-VL are used for model training and evaluation, while the associated textual descriptions are not utilized.

2) FGFD Dataset: The FGFD dataset comprehensively covers diverse farmland landscapes, including plains, plateaus, and hilly regions, providing a broad and challenging benchmark for model evaluation. The dataset is constructed from very-high-resolution (VHR) satellite imagery acquired via the Google Earth platform, with a ground sampling distance (GSD) of 0.3 m. It contains a total of 2606 images, each with a fixed image size of 512×512 pixels, covering an overall area of approximately 70 km^2. To ensure a balanced representation of diverse terrains during model evaluation, a regionwise stratified sampling strategy is adopted. Specifically, images from each geographical landscape are independently partitioned at an 8:1:1 ratio, resulting in exactly 2083 images for training, 263 for validation, and 260 for testing.

3) LoveDA Dataset: The LoveDA dataset comprises 5987 high-resolution optical remote sensing images acquired from Google Earth, spanning three major Chinese cities: Nanjing, Changzhou, and Wuhan. Each image features three spectral bands (RGB) and is annotated with seven distinct land-cover categories: building, road, water, barren land, forest, farmland, and background. The original image dimensions are 1024×1024 pixels, with a spatial resolution of 0.3 m. For the farmland segmentation task, our evaluation is restricted to the rural subset of the LoveDA dataset, which contains a total of 2358 rural images. To adapt the original multiclass dataset to a binary farmland segmentation task, the farmland class is treated as the positive foreground target with pixel value 1, while the other six categories are merged into a single negative background class with pixel value 0. Finally, all images and their corresponding binary masks are downsampled to 512×512 pixels during preprocessing.

B. Experimental Settings

1) Implementation Details: The proposed model is implemented using the PyTorch framework, and all experiments are conducted on a computing platform equipped with two NVIDIA GeForce RTX 4090 GPUs, each with 24 GB of memory. The ViT-B variant of SAM is adopted as the backbone network. During training, the batch size is set to 4, and the model is trained for 200 epochs. For optimization, the AdamW optimizer is employed with an initial learning rate of $2e^{-4}$, which is gradually decayed using a cosine annealing schedule. To ensure stable convergence, all parameters and hyperparameters are carefully tuned in accordance with common practices in related studies. To ensure a strictly fair comparison, all models were reimplemented using their official source codes and evaluated under identical experimental settings, including the same data splits, unified preprocessing pipelines, and standardized training configurations.

2) Evaluation Metrics: Four widely used metrics in agricultural land segmentation are employed to assess the final model accuracy: Dice coefficient (Dice), IoU, mean accuracy (mACC), and recall. Dice measures the similarity between the predicted and ground truth sets. IoU evaluates the average ratio of the intersection to the union of predicted and ground truth regions, serving as a standard metric in semantic segmentation tasks. The mACC metric denotes the average classification accuracy over all classes. Recall measures the proportion of correctly identified positive samples by the model. The

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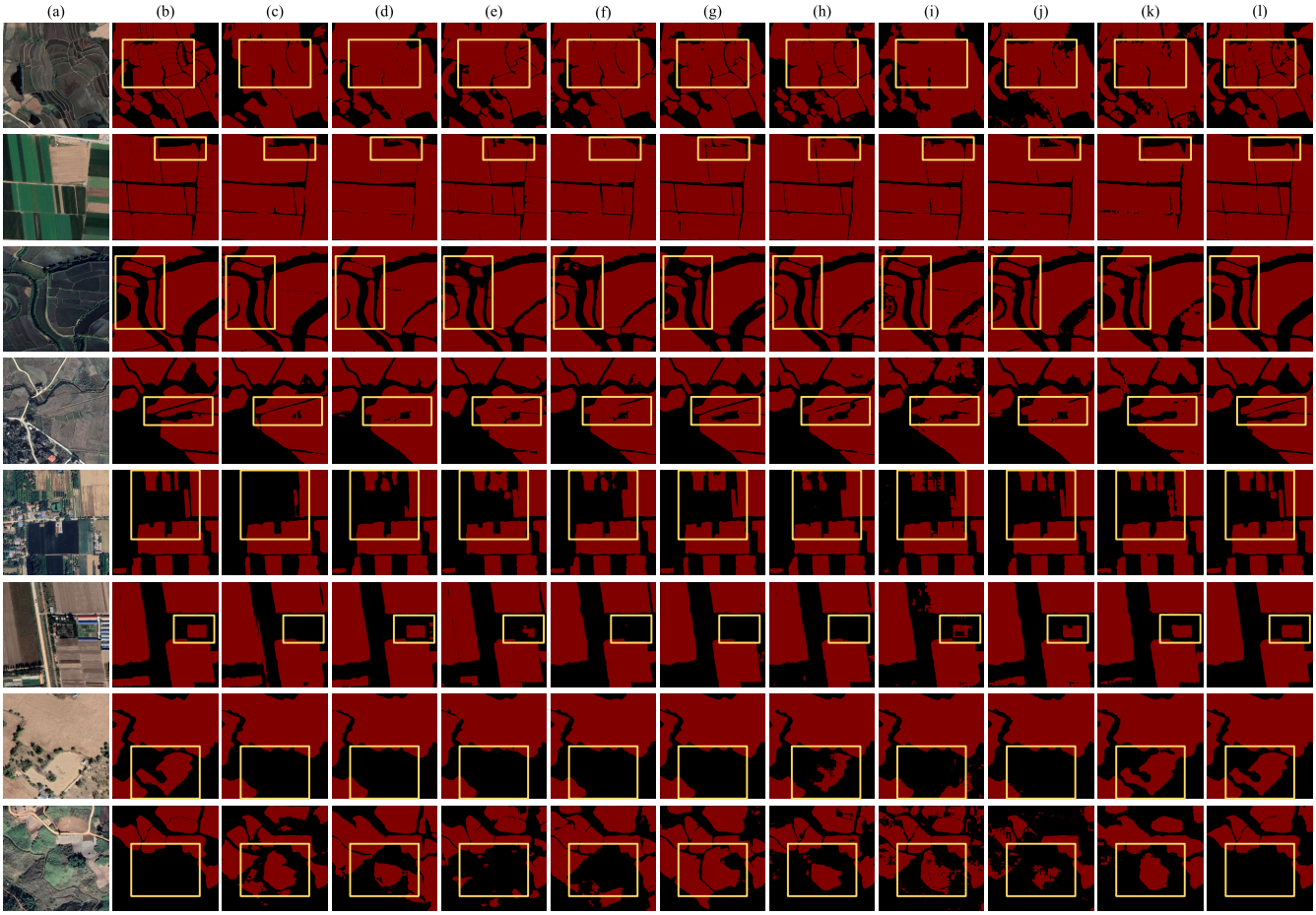


Fig. 6. Visual comparison on the FarmSeg-VL dataset. (a) Input. (b) Truth. (c) Mask2Former. (d) DDRNet. (e) BsiNet. (f) SBDNet. (g) HBGNet. (h) CTHBNet. (i) SAM. (j) RSAM-seg. (k) FSVLM. (l) Ours.

definitions of these metrics are as follows:

$$\text{Dice} = \frac{2 \times \text{TP}}{2 \times \text{TP} + \text{FP} + \text{FN}} \tag{28}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{29}$$

$$\text{mACC} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{30}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{31}$$

where TP, FP, TN, and FN represent the true positive count, false positive count, true negative count, and false negative count, respectively.

In addition, boundary IoU (BIoU) and boundary $F1$-score (BF1) are introduced in the comparative experiments to further evaluate boundary-level segmentation quality across different methods. BIoU focuses on the overlap between predicted and ground truth boundaries, while BF1 quantifies the balance between boundary precision and recall.

C. Comparative Experiments

To thoroughly evaluate the effectiveness of the proposed method, we conducted systematic comparisons on the FarmSeg-VL, FGFD, and LoveDA datasets. The method was compared with three categories of advanced baseline approaches: label-based deep learning methods (Mask2Former [63], DDRNet [64]), multitask joint representation learning methods (BsiNet [41], SBDNet [65], HBGNet [66], CTHBNet [67]), and SAM-based improved methods (SAM [24], RSAM-Seg [52], FSVLM [60]).

1) Results on the FarmSeg-VL Dataset: Fig. 6 provides a qualitative visual comparison between SMPPL and state-of-the-art (SOTA) methods on the FarmSeg-VL test set. In the mountainous terraced field scene depicted in the first row, the farmland exhibits complex curved structures with blurred adjacent boundaries. Mask2Former, DDRNet, and BsiNet show varying degrees of boundary adhesion and structural fragmentation in this region, while RSAM-Seg and FSVLM improve overall coherence but still fail to fully capture the curved boundaries. In contrast, SMPPL more accurately fits the true contours, maintaining continuous and natural curved structures within the yellow boxed area, demonstrating a strong modeling capability for complex geometric shapes. In the fragmented farmland and mixed background scene shown in the fourth row, interference factors such as buildings and roads pose higher demands on model discrimination. Most comparison methods show varying degrees of misclassification in this scenario, mistakenly predicting some background areas as farmland. In contrast, SMPPL maintains stable predictions amid complex backgrounds, demonstrating stronger

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TABLE III
PERFORMANCE COMPARISON OF DIFFERENT MODELS ON THE FARMSEG-VL DATASET

Methods	Dice(%) ↑	IoU(%) ↑	mACC(%) ↑	Recall(%) ↑	BIoU(%)↑	BF1(%) ↑
Mask2Former [63]	86.93	79.68	89.60	88.69	50.86	65.98
DDRNet [64]	88.86	82.08	90.07	90.23	49.54	64.62
BsiNet [41]	85.48	77.83	88.45	86.67	51.74	66.36
SBDNet [65]	88.19	81.04	90.20	90.75	52.56	67.46
HBGNet [66]	90.36	84.10	92.16	90.01	61.62	74.87
CTHBNet [67]	88.12	82.15	90.71	91.39	55.60	69.71
SAM [24]	83.17	74.77	86.97	87.14	45.49	60.84
RSAM-Seg [52]	89.16	82.71	91.45	91.30	53.93	68.54
FSVLM [60]	90.08	82.23	91.21	90.64	55.72	69.53
SMPPL (ours)	**91.45**	**86.22**	**93.01**	**93.20**	**63.53**	**76.37**

semantic discrimination ability and robustness. Overall, traditional label-based deep learning methods, such as Mask2Former and DDRNet, can roughly locate major farmland areas, but they often suffer from severe boundary degradation. As shown in the yellow boxes, these methods struggle to maintain structural continuity, especially in challenging regions with fragmented parcels, narrow boundaries, or irregular field shapes. Although models such as HBGNet and RSAM-Seg improve the segmentation quality to some extent, they still show broken narrow parcels, missed small regions, and local misclassifications under these challenging conditions. In contrast, the segmentation mask generated by our proposed SMPPL model Fig. 6(l) closely matches the ground truth Fig. 6(b). SMPPL shows clear advantages in preserving sharp farmland boundaries and maintaining parcel-level structural completeness while effectively reducing the omission and adhesion errors commonly observed in other methods.

SMPPL achieved the best performance across all four evaluation metrics, as shown in Table III. Compared with HBGNet, SMPPL improved the IoU by 2.12% and the recall by 3.19%, demonstrating clear advantages in maintaining farmland region integrity and boundary restoration. In addition to the commonly used region-level metrics, we further introduced boundary-oriented evaluation metrics, including BIoU and BF1, to more comprehensively assess the ability of different methods to delineate farmland parcel contours. The quantitative results show that SMPPL also achieves the best performance in terms of BIoU and BF1. Compared with the strongest competing method, HBGNet, SMPPL improves BIoU and BF1 by 1.91% and 1.50%, respectively. These improvements demonstrate that SMPPL not only enhances region-level segmentation accuracy but also substantially improves boundary-level representation. It is noteworthy that the original SAM model achieved an IoU of only 74.77% on the FarmSeg-VL dataset, highlighting its limited domain adaptation capability for remote sensing farmland scenes. By introducing structural guidance and semantic prototype constraints, SMPPL improved the IoU by 11.45% compared with SAM, demonstrating the proposed method's significant adaptability to remote sensing farmland scenarios. Compared with other SAM-based improvements, such as RSAM-Seg and FSVLM, SMPPL continues to exhibit consistent performance gains. These superior results indicate that by combining high-frequency structure awareness with multiprototype learning, our approach effectively enhances the model's discriminative capability for farmland.

2) Results on the FGFD Dataset: To verify the actual segmentation performance of each method, we selected representative complex farmland scenes from the FGFD test set for visual comparison, as shown in Fig. 7. In the boundary regions highlighted by the yellow boxes, Mask2Former and DDRNet often fail to detect narrow ridges, leading to frequent breaks in segmentation or the merging of neighboring parcels. SAM and RSAM-Seg tend to produce segmentation results with irregular noise along the boundaries or incomplete internal structures. In contrast, SMPPL accurately delineates extremely thin linear boundaries while maintaining the topological integrity of the parcels. This excellent performance is attributed to our proposed structure-aware mechanism, which explicitly models the spatial continuity and boundary morphology of farmland, enhancing the model's sensitivity to high-frequency spatial structures. As a result, the predictions better reflect the true shape and local boundaries of the parcels.

To further validate the model's stability across different data distributions, comparative experiments were conducted on the FGFD dataset, with results shown in Table IV. SMPPL achieved Dice and IoU scores of 90.25% and 84.18%, respectively, significantly outperforming most baseline methods and demonstrating strong robustness and consistency. For the mACC metric, HBGNet obtained the highest value of 84.47%, slightly higher than that of SMPPL, but its recall and IoU were both lower than those of SMPPL. This indicates that while HBGNet performs well in overall pixel classification accuracy, it still has limitations in capturing fine farmland structures and maintaining regional integrity. In contrast, SMPPL achieved the highest recall score of 94.37%, indicating a stronger perception ability in detecting small targets, blurred boundaries, and complex scenes, better preserving farmland topology and spatial continuity. In terms of the two boundary-related metrics, BIoU and BF1, SMPPL also achieves the best results, reaching 55.59% and 69.64%, respectively. These improvements suggest that the proposed method is more effective in delineating narrow ridges, irregular field edges,

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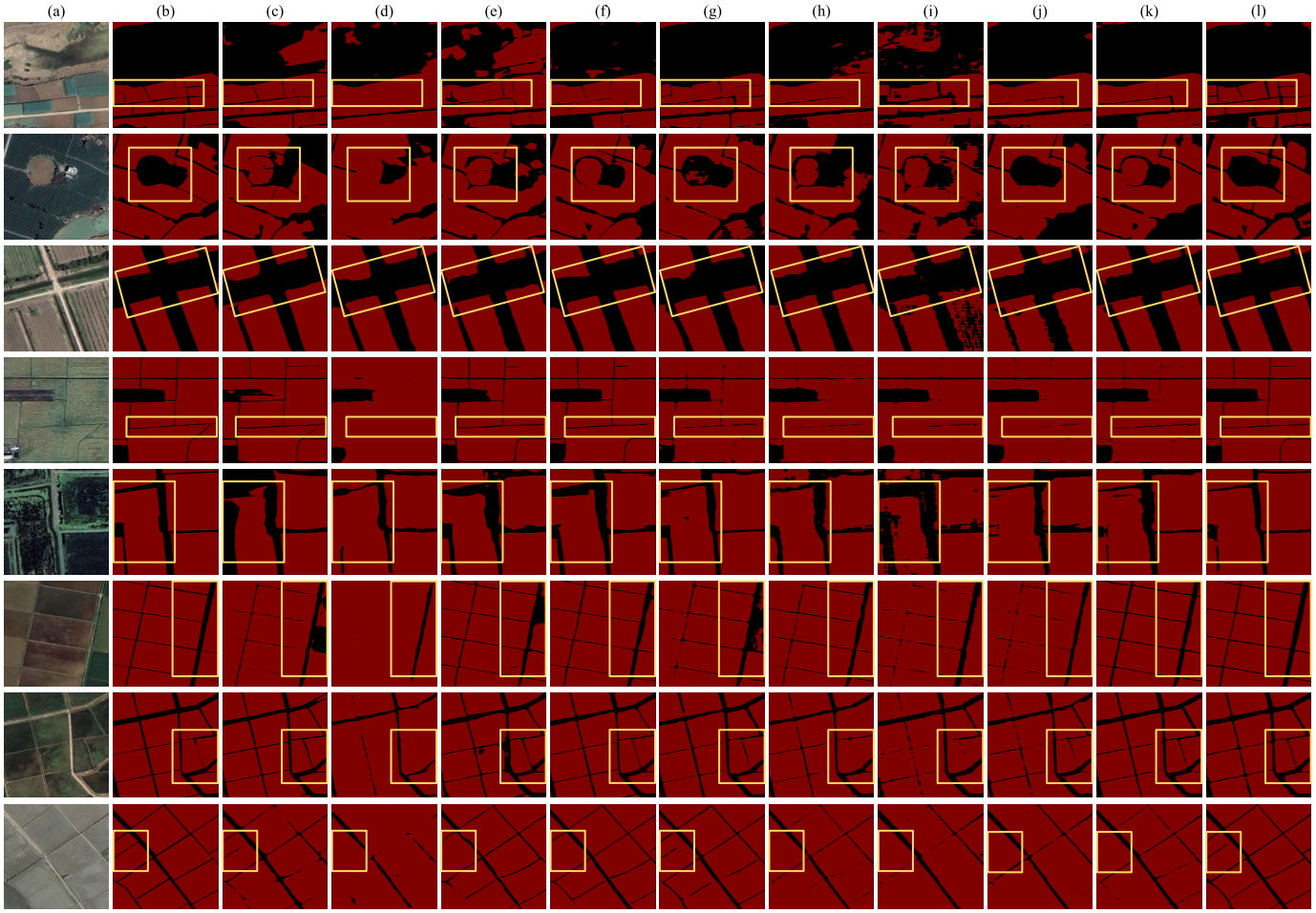


Fig. 7. Visual comparison on the FGFD dataset. (a) Input. (b) Truth. (c) Mask2Former. (d) DDRNet. (e) BsiNet. (f) SBDNet. (g) HBGNet. (h) CTHBNet. (i) SAM. (j) RSAM-seg. (k) FSVLM. (l) Ours.

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT MODELS ON THE FGFD DATASET

Methods	Dice(%) ↑	IoU(%) ↑	mACC(%) ↑	Recall(%) ↑	BIoU(%) ↑	BF1(%) ↑
Mask2Former [63]	83.24	75.52	79.01	87.23	35.84	50.24
DDRNet [64]	86.96	82.04	80.95	90.10	30.49	45.08
BsiNet [41]	86.71	79.45	82.17	89.15	49.79	65.23
SBDNet [65]	88.45	81.84	82.65	91.98	49.30	64.44
HBGNet [66]	89.13	82.49	**84.47**	92.97	51.86	66.28
CTHBNet [67]	88.21	81.29	81.06	90.73	48.53	62.48
SAM [24]	83.40	75.25	78.07	80.31	37.83	53.15
RSAM-Seg [52]	88.52	81.86	81.76	92.89	42.88	58.48
FSVLM [60]	89.01	81.42	81.25	91.68	45.89	60.71
SMPPL (ours)	**90.25**	**84.18**	83.73	**94.37**	**55.59**	**69.64**

and fragmented parcel boundaries. Overall, the results on the FGFD dataset further confirm that SMPPL maintains stable and superior segmentation performance across varied data distributions and complex scenarios.

D. Generalization Performance of SMPPL

Farmland spatial structures and distributions exhibit significant irregularity and specificity, posing substantial challenges to the generalization capability of segmentation models. To evaluate the generalization performance of the proposed SMPPL, we conducted a quantitative analysis of its predictions on the challenging LoveDA dataset. Specifically, all methods were first trained under a unified setting on the FarmSeg-VL dataset. Subsequently, the LoveDA dataset was used as an independent testing benchmark, where the trained models were directly transferred to a new geographic region for evaluation without additional fine-tuning. For this experiment, only the

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TABLE V
COMPARISON OF GENERALIZATION PERFORMANCE ACROSS DIFFERENT MODELS ON THE LOVEDA DATASET

Methods	Dice(%) ↑	IoU(%) ↑	mACC(%) ↑	Recall(%) ↑	BIoU(%) ↑	BF1(%) ↑
Mask2Former [63]	62.44	51.63	72.45	63.32	26.75	42.68
DDRNet [64]	58.74	47.32	61.89	58.53	23.56	36.53
BsiNet [41]	58.99	48.15	68.42	60.03	28.94	34.28
SBDNet [65]	59.15	48.93	70.15	58.28	30.29	39.04
HBGNet [66]	61.10	50.47	72.31	57.04	32.16	44.76
CTHBNet [67]	60.40	49.53	70.39	62.59	30.26	40.12
SAM [24]	64.69	54.63	75.03	63.92	26.38	39.62
RSAM-Seg [52]	61.78	50.89	71.21	65.05	25.64	40.36
FSVLM [60]	62.24	51.56	70.78	62.12	28.22	41.84
SMPPL (ours)	**67.66**	**56.98**	**78.51**	**67.75**	**35.48**	**50.24**

TABLE VI
ABLATION STUDY RESULTS OF THE SMPPL

Model Configuration	SFEM	SSCM	MPGPM	Dice(%) ↑	IoU(%) ↑	mACC(%) ↑	Recall(%) ↑
S0	×	×	×	86.16	80.60	88.54	90.30
S1	✓	×	×	88.86	82.75	90.72	91.45
S2	×	×	✓	89.86	83.33	90.87	91.78
S3	✓	✓	×	90.23	84.06	91.58	92.52
S4	✓	×	✓	91.24	85.56	92.80	93.05
SMPPL (ours)	✓	✓	✓	**91.45**	**86.22**	**93.01**	**93.20**

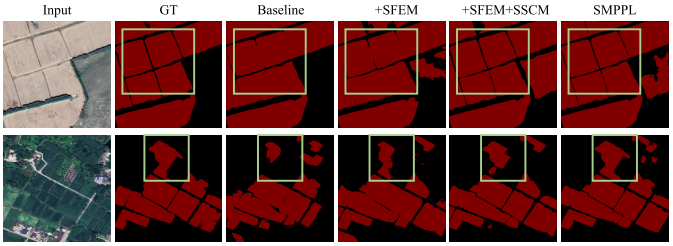


Fig. 8. Visual comparison of baseline with progressively added SFEM, SSCM, and MPGPM.

TABLE VII
COMPARISON OF DIFFERENT FEATURE FUSION STRATEGIES

Model Configuration	Dice(%) ↑	IoU(%) ↑	mACC(%) ↑	Recall(%) ↑
w/o SSCM	91.24	85.56	92.80	93.05
w/o SSCM + CW	91.31	85.60	92.88	92.85
w/o SSCM + SA	91.38	85.72	92.94	93.08
SMPPL (ours)	**91.45**	**86.22**	**93.01**	**93.20**

2358 rural samples from the LoveDA dataset were selected for evaluation. The corresponding results are presented in Table V. The experimental results demonstrate that the proposed method maintains stable and superior segmentation performance even in completely unseen geographic scenarios and overall outperforms the other comparative methods by a clear margin.

From the perspective of semantic consistency, the prototype prompting mechanism enhances the consistency of intra-class feature representations by constructing stable class-level semantic prototypes, thereby enabling the model to maintain robust discriminative capability in complex backgrounds and cross-regional scenarios. This advantage becomes particularly evident in the cross-dataset experiments: even when the testing regions exhibit significant differences from the training data in terms of imaging conditions and geographic distributions, SMPPL is still able to maintain high segmentation accuracy.

E. Ablation Experiment

To analyze the contribution of each component to the overall performance of the model, systematic ablation experiments were conducted on the FarmSeg-VL dataset. The results are presented in Table VI. Specifically, by progressively introducing SFEM, SSCM, and MPGPM, with (w/o) indicating the removal of a component and (+) indicating its inclusion in the experiments, we comparatively analyze the impact of each module on the performance of farmland segmentation.

1) Impact of SFEM: To analyze the effect of SFEM, we compare the experimental results obtained with and without this component. When only MPGPM is employed without incorporating SFEM, the model achieves Dice, IoU, mACC, and Recall scores of 89.86%, 83.33%, 90.87%, and 91.78%, respectively. After introducing SFEM, the model performance improves significantly, as shown in Fig. 8. These results indicate that SFEM effectively enhances the model's ability to represent the spatial structural information of farmland. By modeling multiscale structural features, the module strengthens the representation of farmland boundaries and morphological patterns, enabling the encoder to learn more stable and discriminative feature representations, thereby improving the overall segmentation accuracy.

2) Impact of SSCM: To validate the effectiveness of the proposed SSCM, we conduct replacement experiments within the SMPPL method using different feature fusion strategies. Specifically, after removing SSCM, we replace it with commonly used feature reweighting mechanisms, including channelwise weighting (CW) and SA, and compare them with the complete model. The corresponding experimental results are presented in Table VII. These results indicate that the SSCM effectively enhances the consistency between structural and semantic representations. As the visualization results in

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TABLE VIII
PERFORMANCE COMPARISON UNDER DIFFERENT OUTPUT PROJECTION MATRIX C STRATEGIES

Strategy	Dice(%) ↑	IoU(%) ↑	mACC(%) ↑	Recall(%) ↑
No Shared	90.63	84.82	91.51	91.85
SharedC	**91.45**	**86.22**	**93.01**	**93.20**

TABLE IX
PERFORMANCE COMPARISON WITH DIFFERENT NUMBERS OF PROTOTYPES

Numbers of Prototypes	Dice(%) ↑	IoU(%) ↑	mACC(%) ↑	Recall(%) ↑
3	91.31	85.62	92.25	91.83
4	91.34	85.67	92.08	92.15
5	**91.45**	**86.22**	**93.01**	**93.20**
6	91.39	85.81	92.13	92.18

Fig. 8 demonstrate, integrating the SSCM further enhances segmentation integrity. By introducing structural–semantic constraints during feature learning, the module reduces semantic inconsistency within farmland regions and enhances the model's capability for boundary refinement.

To further validate the effectiveness of the shared output projection strategy in SSCM, we conducted an additional comparative experiment by replacing the shared projection matrix C with two independent projection matrices for the semantic and structural branches, respectively. The corresponding results are presented in Table VIII. It can be observed that the shared projection strategy consistently achieves superior performance across all evaluation metrics, improving Dice from 90.63% to 91.45%, IoU from 84.82% to 86.22%, mACC from 91.51% to 93.01%, and recall from 91.85% to 93.20%. These results demonstrate that introducing a shared output projection matrix effectively enhances the consistency between semantic and structural representations during the state-space transformation process. Unlike independent projections, the shared projection constrains both branches to be mapped into a unified feature subspace, thereby reducing semantic divergence between heterogeneous features and improving cross-branch feature alignment. As a result, the fused representations become more discriminative and structurally coherent, which is particularly beneficial for preserving farmland boundary continuity and regional integrity in complex remote sensing scenes.

3) Impact of MPGPM: To determine the optimal configuration for the MPGPM, we conduct a parameter ablation study on the number of prototypes N, as shown in Table IX. We evaluate the segmentation performance by varying N from 3 to 6. The results demonstrate that the model achieves its best performance at $N = 5$, yielding a top Dice score of 91.45% and an IoU of 86.22%. When N is less than 5, the prototypes are insufficient to fully capture the diverse structural patterns of farmland regions. Conversely, increasing N to 6 leads to a slight performance degradation, indicating that excessive prototypes may introduce computational redundancy or cause overfitting to local noise. Therefore, $N = 5$ is adopted as the optimal setting to balance representation capacity and model efficiency.

TABLE X
ABLATION STUDY RESULTS OF THE MPGPM

DPG	SPG	Dice(%) ↑	IoU(%) ↑	mACC(%) ↑	Recall(%) ↑
×	×	90.23	84.06	91.58	92.52
×	✓	89.78	84.82	91.86	92.26
✓	×	90.64	85.15	92.05	92.83
✓	✓	**91.45**	**86.22**	**93.01**	**93.20**

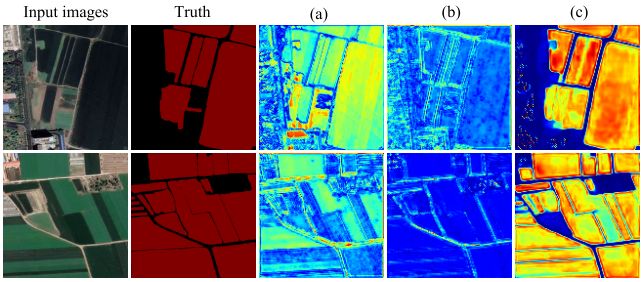


Fig. 9. Visualization of intermediate feature responses of the proposed SMPPL. (a)–(c) Feature responses from the SAM image encoder, the structure-aware encoder, and the semantic–structure fused representation, respectively.

Furthermore, to systematically evaluate the effectiveness of MPGPM, we conduct a progressive ablation study to analyze the individual and collaborative contributions of its core subcomponents: DPG and SPG. The quantitative results are reported in Table X. When operating without both subcomponents, the baseline yields an IoU of 84.06% and a Dice score of 90.23%. By incorporating only the SPG module, the model leverages semantic prompts to slightly boost the IoU to 84.82%. Conversely, integrating only the DPG module injects pixel-level dense spatial context, leading to a more noticeable performance improvement, with the IoU and Dice scores increasing to 85.15% and 90.64%, respectively. Ultimately, simultaneously coupling both DPG and SPG achieves the best performance across all evaluated metrics, elevating the Dice, IoU, mACC, and recall scores to 91.45%, 86.22%, 93.01%, and 93.20%, respectively. The visualization results in Fig. 8 visually demonstrate the effectiveness of MPGPM in addressing high intraclass variance. By constructing multiple prototype-based semantic representations, the module enhances intraclass feature consistency and effectively suppresses background noise interference during segmentation.

4) Visualization of Intermediate Features: To further improve the explainability of the proposed SMPPL, we visualize representative intermediate feature responses, as shown in Fig. 9. In the early feature response shown in Fig. 9(a), the model mainly focuses on local texture patterns and low-level spatial variations. After introducing structure-aware modeling, the feature map in Fig. 9(b) shows clearer responses around farmland boundaries and internal parcel

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TABLE XI
PERFORMANCE COMPARISON WITH DIFFERENT LOSS WEIGHT COEFFICIENT

λ_3	Dice(%) ↑	IoU(%) ↑	mACC(%) ↑	Recall(%) ↑
0.2	90.89	85.18	92.30	91.66
0.5	**91.45**	**86.22**	**93.01**	**93.20**
0.7	91.06	85.39	92.81	92.88
1.0	90.46	84.49	91.63	92.35

structures. Thin ridges, field edges, and adjacent parcel separations become more distinguishable, indicating that the structure-aware branch helps the model capture high-frequency boundary cues. The feature response in Fig. 9(c) further shows a stronger focus on complete farmland parcels, producing clearer parcel-level responses that are better aligned with the ground truth masks. This indicates that the semantic–structure coordination and prompt-guided refinement process help the model integrate boundary details with region-level semantic information. Overall, the visualization results demonstrate that SMPPL gradually shifts its attention from low-level texture responses to structure-aware and semantically consistent farmland representations.

5) Sensitivity Analysis of Loss Weighting Coefficients: To further validate the rationality of the loss weighting coefficients in the proposed joint optimization objective, we conducted a sensitivity analysis on the boundary-aware loss weight λ_3, while fixing $\lambda_1 = \lambda_2 = 1.0$. As shown in Table XI, the model achieves the best overall performance when $\lambda_3 = 0.5$, obtaining Dice, IoU, mACC, and Recall scores of 91.45%, 86.22%, 93.01%, and 93.20%, respectively. When λ_3 is set to a relatively small value, the contribution of the boundary-aware optimization becomes insufficient, limiting the model's ability to capture fine-grained parcel boundaries and complex geometric structures. Consequently, the segmentation accuracy decreases, particularly in terms of IoU and recall. Conversely, excessively increasing the boundary-aware weight, such as to $\lambda_3 = 0.7$ or $\lambda_3 = 1.0$, causes the model to overemphasize local boundary details, which weakens global regional consistency and introduces instability in semantic region prediction. This results in a gradual decline in overall segmentation performance. These results indicate that an appropriate balance between regional supervision and boundary-aware optimization is crucial for farmland segmentation. The selected setting, $\lambda_1 = 1.0$, $\lambda_2 = 1.0$, and $\lambda_3 = 0.5$, achieves the most effective tradeoff between structural boundary preservation and semantic region consistency, thereby yielding the most stable and accurate segmentation results.

6) Efficiency Analysis and Limitations: To evaluate the practical applicability of the proposed SMPPL framework, we evaluate the efficiency of SMPPL in terms of parameters (params), FLOPs, and frames per second (FPS), as shown in Table XII. In the params column, the values before and after the slash denote trainable params and total params, respectively. In terms of trainable params, SMPPL contains 16.69 M trainable params, which is lower than that of Mask2Former, DDRNet, SBDNet, HBGNet, and CTHBNet. This indicates

TABLE XII
EFFICIENCY COMPARISON OF DIFFERENT METHODS

Method	Trainable/Total Params (M) ↓	FLOPs (G) ↓	FPS ↑
Mask2Former [63]	43.98 / 43.98	25.70	56.52
DDRNet [64]	20.30 / 20.30	**18.60**	**314.04**
BsiNet [41]	7.85 / **7.85**	53.13	302.26
SBDNet [65]	36.63 / 36.63	104.16	42.68
HBGNet [66]	30.45 / 30.45	231.92	24.57
CTHBNet [67]	74.08 / 74.08	94.79	42.55
SAM [24]	**4.06** / 91.34	103.18	70.22
RSAM-Seg [52]	14.77 / 102.05	112.45	58.53
SMPPL (ours)	16.69 / 103.97	131.53	28.15

that the proposed method still maintains a relatively PEFT strategy by freezing most params of the SAM backbone and only optimizing the newly introduced lightweight modules. However, SMPPL has relatively high total params and computational complexity. Specifically, SMPPL contains 103.97 M total params and requires 131.53 G FLOPs. These results indicate that although the proposed method achieves superior segmentation accuracy, especially in boundary-related metrics and complex farmland scenes, it still introduces additional computational cost. Therefore, SMPPL presents a tradeoff between segmentation accuracy and computational efficiency. Compared with lightweight networks such as DDRNet and BsiNet, SMPPL is less advantageous in inference speed. Nevertheless, the proposed method achieves significantly better segmentation performance, particularly in IoU, Recall, BIoU, and BF1, demonstrating that the additional computational cost contributes to more accurate farmland region extraction and boundary delineation. This tradeoff is acceptable in scenarios where segmentation accuracy and parcel boundary integrity are more important than real-time inference speed, such as high-precision farmland mapping and agricultural resource monitoring.

The main limitation of the current framework lies in its relatively high FLOPs requirement for ultralarge-scale imagery. In future work, we will focus on improving the efficiency of SMPPL by exploring lightweight SAM backbones, model pruning, knowledge distillation, and more efficient structural feature modeling strategies. In addition, designing a more compact prototype prompt mechanism and reducing redundant feature interactions will be important directions for achieving a better balance between segmentation accuracy and computational efficiency.

V. CONCLUSION

This article proposes an SMPPL method to address the challenges of complex parcel structures and large intraclass variability in remote sensing farmland segmentation. By introducing a parallel structure-aware encoder and SSCM, the method effectively addresses the limitations of SAM in fine-grained boundary recovery. Furthermore, we design an MPGPM. It updates the prototypes dynamically using a class-attention mechanism and generates spatially aligned dense prompts along with globally discriminative sparse prompt

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embeddings, facilitating deep interaction between semantic prompts and spatial features. Experimental results on multiple public farmland remote sensing datasets demonstrate that the proposed method consistently outperforms existing approaches, particularly in complex boundary reconstruction, regional consistency preservation, and cross-dataset generalization.

Despite its promising performance in farmland segmentation, the proposed method presents two main limitations. First, the proposed architecture introduces additional computational overhead, which may constrain scalability and inference efficiency for ultralarge-scale imagery. Second, relying primarily on RGB optical data and Chinese agricultural scenes restricts the comprehensive evaluation of its global generalization. To address these constraints, our future work will focus on two key directions. First, we will improve computational efficiency and deployment feasibility by optimizing the architecture, including the integration of next-generation foundation models and more efficient Mamba variants. Second, to enhance model robustness and generalization under complex environments, we will explore multisensor data fusion by integrating SAR and optical imagery.

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